anderson®
Diagnostics & Labs
Adding Health to Life...

Anderson Diagnostic Services Pvt. Ltd.
No. 26, Rajarathinam Street, Kilpauk, Chennai - 600 010.
Helpline No: 044-4353 9444 / 99620 00048
Email: enquiries@andersondiagnostics.com
www.andersondiagnostics.com

Dementia NGS Sequencing Analysis

Patient name: XX	**PIN:** XX
Age: 74 Years	**Sex:** Male
Hospital/Clinic: XX	**Sample Number:** XX
Referring Clinician: XX	**Sample Collection Date:** NA
Specimen: Peripheral Blood	**Sample Received Date:** NA
	Report Date: NA

Clinical History:

Proband, XX is suspected to be affected with Alzheimer's disease and dementia. Proband, XX has been evaluated for pathogenic variations.

Results

Pathogenic variant was identified related to the given phenotype

List of significant variants identified:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
GRN (+)	c.560del **(p.Leu189ArgfsTer9)**	Heterozygous	Exon 6	Frontotemporal dementia 2 / Aphasia, primary progressive **(OMIM#607485)**	Autosomal Dominant / Autosomal Recessive	**Pathogenic (PVS1, PM2_Supp, PP5)**

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].
No other variant that warrants to be reported for the given clinical indication was identified

Variant Interpretation:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
GRN (+)	c.560del **(p.Leu189ArgfsTer9)**	Autosomal Dominant / Autosomal Recessive	Frontotemporal dementia 2 / Aphasia, primary progressive **(OMIM#607485)**	PVS1, PM2_Supp, PP5	**Pathogenic**
Region	**Allele Status**	**Sequence Ontology**	**Variant Frequency**	**Allelic Read Depths**	
Exon 6	Heterozygous	Frameshift	0.00006% frequency observed in Annotated gnomAD v4 All	Ref(T): 60, Alt(-): 60, VAF: 50.00%	
***In silico* Prediction Tools**				**Conservation Status**	
SIFT: **-** LRT: **-** PolyPhen2: **-** CADD: **T** MT2: **D**				PhyloP: **-** GERP++: **-**	
Genomic Position			**Clinical & Literature Evidences**		
Chr17:NC_000017.11:g.44350538delT (NM_024844.5)			ClinVar: 599615 PubMed: 36970912		

Abbreviation: D: Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting;

Clinical and Literature Evidence: This disease has been reported as pathogenic in ClinVar database. This variant has been previously reported in patients affected with Frontotemporal Dementia.

OMIM Phenotype: Frontotemporal dementia 2 / Aphasia, primary progressive (OMIM#607485) is caused by mutation in the GRN gene (OMIM*138945) Frontotemporal dementia-2 (FTD2) is a neurodegenerative disorder characterized by adult onset of progressive cognitive decline which shows variable phenotypic expression but most commonly presents as behavioral changes with executive dysfunction (bvFTD) and/or language deterioration. The age at onset for individuals with heterozygous GRN mutations ranges from 40 to 85 years. Gestural apraxia, parkinsonism, visual loss, and visual hallucinations are present in 25 to 40% of patients. Other variations of the phenotype have been referred to as 'dysphasic disinhibition dementia,' 'primary progressive aphasia' (PPA), or 'corticobasal degeneration.' Some patients may present with a clinical diagnosis of Alzheimer disease or Parkinson disease, features of which are part of the phenotypic spectrum of this disorder. Most GRN mutations are null and lead to haploinsufficiency. Rare individuals with biallelic, likely hypomorphic mutations have been reported. These disease follow both autosomal dominant and autosomal recessive pattern of inheritance [2].

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].
No other variant that warrants to be reported for the given clinical indication was identified

Recommendations:

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Anderson Diagnostic Services Pvt. Ltd.
No. 26, Rajarathinam Street, Kilpauk, Chennai - 600 010.
Helpline No: 044-4353 9444 / 99620 00048
Email: enquiries@andersondiagnostics.com
www.andersondiagnostics.com

Methodology:

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re- alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

Sequence data attributes:

Total Read Generated	**17.44 Gb**
Data ≥ Q30	**96.01%**

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients’ symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer:

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.

Anderson Diagnostic Services Pvt. Ltd.
No. 26, Rajarathinam Street, Kilpauk, Chennai - 600 010.
Helpline No: 044-4353 9444 / 99620 00048
Email: enquiries@andersondiagnostics.com
www.andersondiagnostics.com

- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions. The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1, SMN2, HBA1, HBA2, PMS2, GBA,* and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

References:

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Miller, David T., et al. "ACMG SF v3. 1 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in Medicine 24.7 (2022): 1407-1414.

This report has been reviewed and approved by:

Sivasankar.S, Ph.D Molecular Biologist	Muthukumaran. S, Ph.D Clinical Bioinformatician	Sachin. D.Honguntikar, Ph.D, Molecular Geneticist	Dr. G. Suriyakumar Director

Anderson Diagnostic Services Pvt. Ltd.
No. 26, Rajarathinam Street, Kilpauk, Chennai - 600 010.
Helpline No: 044-4353 9444 / 99620 00048
Email: enquiries@andersondiagnostics.com
www.andersondiagnostics.com

Appendix 1: Gene Coverage Indication Based Analysis:

Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered
A2M	100.0	*AARS1*	100.0	*ABCA7*	100.0	*ABCB7*	100.0
ABCD1	100.0	*ACE*	100.0	*ADA2*	100.0	*ADAM10*	100.0
ADH1C	100.0	*ALDH18A1*	100.0	*ALS2*	100.0	*ANG*	100.0
ANXA11	100.0	*AP4E1*	100.0	*APOE*	100.0	*APP*	100.0
APTX	100.0	*ARSA*	100.0	*ASAH1*	100.0	*ATN1*	100.0
ATP13A2	100.0	*ATP1A1*	100.0	*ATP6V0A2*	100.0	*ATP6V1A*	100.0
ATP6V1E1	100.0	*ATP7B*	100.0	*ATRX*	100.0	*ATXN10*	100.0
ATXN2	100.0	*ATXN3*	100.0	*AUH*	100.0	*BIN1*	100.0
BLTP1	100.0	*BRAF*	100.0	*BVES*	100.0	*C19orf12*	100.0
C9orf72	100.0	*CACNA1G*	100.0	*CBLIF*	100.0	*CCNF*	100.0
CD33	100.0	*CDH23*	100.0	*CERS1*	100.0	*CFAP410*	100.0
CFAP43	100.0	*CHCHD10*	100.0	*CHCHD2*	100.0	*CHMP2B*	100.0
CISD2	100.0	*CLN3*	100.0	*CLN6*	100.0	*CLN8*	100.0
CLU	100.0	*CMPK2*	100.0	*COL4A1*	100.0	*COL4A2*	100.0
COQ7	100.0	*CP*	100.0	*CPT1C*	100.0	*CSF1R*	100.0
CST3	100.0	*CSTB*	100.0	*CTSF*	100.0	*CUBN*	100.0
CYLD	100.0	*CYP27A1*	100.0	*CYP2D6*	100.0	*DAO*	100.0
DCTN1	100.0	*DGUOK*	100.0	*DLAT*	100.0	*DNAJC13*	100.0
DNAJC5	100.0	*DNAJC6*	100.0	*DNM1L*	100.0	*DNMT1*	100.0
EIF4G1	100.0	*EPM2A*	100.0	*ERBB4*	100.0	*ERCC4*	100.0
ERCC8	100.0	*FBXO7*	100.0	*FIG4*	100.0	*FMR1*	100.0
FTL	100.0	*FUS*	100.0	*GALC*	100.0	*GBA*	100.0
GBA2	100.0	*GBE1*	100.0	*GCDH*	100.0	*GIGYF2*	100.0
GLE1	100.0	*GLT8D1*	100.0	*GM2A*	100.0	*GPRC5B*	100.0
GPX1	100.0	*GRN*	100.0	*HERC2*	100.0	*HEXA*	100.0
HLA-DQB1	100.0	*HMBS*	100.0	*HNRNPA1*	100.0	*HNRNPA2B1*	100.0
HTRA1	100.0	*HTRA2*	100.0	*HTT*	100.0	*IRF6*	100.0
ITM2B	100.0	*JAM2*	100.0	*JPH3*	100.0	*KCTD7*	100.0
LIG3	100.0	*LMNB1*	100.0	*LRRK2*	100.0	*LYST*	100.0
MAP1B	100.0	*MAPT*	100.0	*MATR3*	100.0	*MBTPS2*	100.0
MMACHC	100.0	*MME*	100.0	*MORC2*	100.0	*MPO*	100.0
MS4A4E	100.0	*MS4A6A*	100.0	*MYORG*	100.0	*NDP*	100.0
NDUFA1	100.0	*NEFH*	100.0	*NEK1*	100.0	*NHLRC1*	100.0
NKX2-1	100.0	*NOS3*	100.0	*NOTCH2NLC*	94.05	*NOTCH3*	100.0
NPC1	100.0	*NPC2*	100.0	*NR3C1*	100.0	*NR4A2*	100.0
OPA1	100.0	*OPTN*	100.0	*PANK2*	100.0	*PARK7*	100.0
PDE10A	100.0	*PDGFB*	100.0	*PDGFRB*	100.0	*PFN1*	100.0
PICALM	100.0	*PINK1*	100.0	*PLA2G6*	100.0	*PLAU*	100.0
PLD3	100.0	*PNPLA6*	100.0	*PODXL*	100.0	*POLG*	100.0
PON1	100.0	*PON2*	100.0	*PON3*	100.0	*PPARGC1A*	100.0
PPP2R2B	100.0	*PRDM8*	100.0	*PRDX1*	100.0	*PRICKLE1*	100.0
PRKAR1B	100.0	*PRKN*	100.0	*PRNP*	100.0	*PRPH*	100.0
PSAP	100.0	*PSEN1*	100.0	*PSEN2*	100.0	*RAB39B*	100.0
RNF216	100.0	*ROGDI*	100.0	*RRM2B*	100.0	*SAMD9L*	100.0
SCARB2	100.0	*SDHA*	100.0	*SDHAF1*	100.0	*SDHB*	100.0
SDHD	100.0	*SERPINI1*	100.0	*SETX*	100.0	*SLC13A5*	100.0
SLC20A2	100.0	*SLC52A3*	100.0	*SNCA*	100.0	*SNCAIP*	100.0
SNCB	100.0	*SOD1*	100.0	*SORL1*	100.0	*SPAST*	100.0
SPG11	100.0	*SPG21*	100.0	*SPTLC1*	100.0	*SQSTM1*	100.0
STARD7	100.0	*SYNJ1*	100.0	*TAF15*	100.0	*TARDBP*	100.0

Anderson Diagnostic Services Pvt. Ltd.
No. 26, Rajarathinam Street, Kilpauk, Chennai - 600 010.
Helpline No: 044-4353 9444 / 99620 00048
Email: enquiries@andersondiagnostics.com
www.andersondiagnostics.com

TBK1	100.0	*TBP*	100.0	*TIA1*	100.0	*TIMM8A*	100.0
TMEM106B	100.0	*TOMM40*	100.0	*TP53*	100.0	*TREM2*	100.0
TRPM7	100.0	*TTR*	100.0	*TUBA4A*	100.0	*TUBB4A*	100.0
TUBGCP6	100.0	*TWNK*	100.0	*TYMP*	100.0	*TYROBP*	100.0
UBQLN2	100.0	*UCHL1*	100.0	*UNC13A*	100.0	*USP48*	100.0
USP8	100.0	*VAPB*	100.0	*VCP*	100.0	*VPS13A*	100.0
VPS13C	100.0	*VPS35*	100.0	*WDR45*	100.0	*WFS1*	100.0
XPR1	100.0	*ZFYVE26*	100.0				